

USBSID-Pico Cynthcart manual

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Disclaimer

I do this stuff in my free time for my enjoyment. Since I like to share my joy in creating this with everyone I try my best to provide a working PCB and Firmware. I am in no way an electronics engineer and can give **no guarantees** that this stuff does not break or damage your hardware, computer, phone, or whatever you try to hook it up to. Be sure to take great care when inserting any real MOS SID chips into the board. While everything has been tested with real chips, this is in no way a guarantee that nothing could go wrong. Use of this board and firmware at your own risk! I am in no way responsible for your damaged hardware. That being said, have fun!

Thank you

First off thanks go out to Paul Slocum for creating Cynthcart and Mario Ballano for creating emudore.

Firmwares with Cynthcart embedded

Starting with firmware version v0.8.0 firmwares ending with `cc` can run Cynthcart. For example `usbsidpico-v1.0_sp_cc.uf2` or `usbsidpico2-rgb-v1.3_sp_cc.uf2` Cynthcart runs through the embedded USBID-Player, so all Cynthcart firmwares `_also` come with the onboard USBID-Player included. See the `USBID-Player-manual` file for more information.

WARNING! Because Cynthcart runs through the embedded USBID-Player, the Pico MCU is overclocked to 250MHz, regardless of Pico type (rp2040 or rp2350). Passive cooling *might/could* be needed.

Usage

To start the emulator and Cynthcart you will need an USB-midi connection to USBID=Pico.

CC messages

85 (0x55) ~ Enable Cynthcart

86 (0x56) ~ Disable Cynthcart

87 (0x57) ~ Reset Cynthcart (**not implemented yet**, currently a no-op)

Cynthcart control

For all other midi CC messages to control Cynthcart please refer to the Cynthcart manual.

Page two in github.com/PaulSlocum/cynthcart/blob/master/manual/cynthcart2.0.1.pdf [cynthcart2.0.1.pdf] or [cynthcart2.0.1_5x7.pdf](#)

On the right side of [cynthcartCommandChart1.pdf](#)

Or on the bottom of [cynthcartCommandChart2.pdf](#)

Development

More information about adorable (emudore-embedded) is available on the [github repo](#)

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